BRIAN NGUYEN
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EDUCATION

The University of Texas at Austin Aug 2021 - May 2025
BSA in Computer Science - GPA: 3.72 | *Certificate in Applied Statistical Modeling*
Relevant Courses: Data Structures, Algorithms and Complexity, Computer Architecture, Operating Systems, Compilers, Concurrency, Machine Learning

EXPERIENCE

Software Engineer | Modern Intelligence Nov 2025 - Present
- Expanded platform support to edge devices by integrating ONNX and TensorRT into Dockerized pipelines, enabling deployment of ML models in resource-constrained environments
- Consolidated separate FFmpeg pipelines for video and KLV data into a single GStreamer pipeline, reducing process overhead and improving streaming efficiency
- Optimized optical flow pipeline to achieve 3x speedup by leveraging GPU acceleration and minimizing CPU-GPU data transfers

Software Engineering Intern | Modern Intelligence Jun 2025 - Oct 2025
- Developed automated tests to validate cursor-on-target accuracy, improving reliability of tracking outputs
- Utilized Docker containers to modularize tightly coupled software and test production-ready deployments

College Readiness Mentor | UT Austin May - Aug 2024, May - Aug 2025
- Guided 250+ incoming students during their transition to college through weekly office hours and online support
- Provided academic resources, tutoring, and general Q&A sessions focused on the College of Natural Sciences

Liberal Arts and Science Academy - *Tutor* | Austin, Texas Feb 2022 - May 2025
- Tutored 10+ students in small group sessions several times a week, supporting their academic growth
- Created personalized study plans and provided tailored assistance to meet individual student needs

Harpak Lab - *Undergraduate Student Researcher* | UT Austin Jan - May 2024
- Conducted genome-wide analyses using Bash, R, and PLINK on the TACC supercomputer system
- Participated in weekly lab meetings to discuss papers on polygenic scores for insights into current advancements

PROJECTS

System Emulator | C
- Emulated hardware elements by writing functions to read/write register values and perform computations (ALU)
- Developed a pipeline implementation for basic instructions from the ARM instruction set architecture

K-Means Clustering | CUDA & C++
- Developed parallel K-Means algorithm on a GPU using CUDA for efficient large-scale data clustering
- Optimized performance by utilizing shared memory and minimizing data transfer overheads

Barnes-Hut Simulation | MPI & C++
- Simulated n-particle gravitational interactions using sequential and parallel algorithms to compare performance
- Achieved speedup and scalability by distributing computations across processes using Message Passing Interface

Two-Phase Commit Protocol | Rust
- Implemented a distributed transaction model with several clients and participants while ensuring consistency
- Utilized inter-process communication channels to manage message-passing and fault tolerance

SKILLS

Java | C/C++ | Python | SQL | HTML/CSS/JS | Spring | Docker | Git | Linux | Bash | Agile/Scrum | CI/CD